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Cognitive behavioral therapy effects on global functioning, domain-specific functioning, and quality of life in adult ADHD: A comprehensive meta-analysis

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ABSTRACT

Background: Cognitive behavioral therapy (CBT) for adult ADHD provides adaptive coping strategies targeting executive dysfunction and emotion regulation, processes underlying functional impairment. However, most research has prioritized symptom reduction over functional outcomes. This meta-analysis quantified CBT effects on global functioning, quality of life (QoL), and domain-specific functioning (social, occupational, educational), while examining moderators and the role of ADHD symptom change.

Method: We included 70 studies (5152 participants): 38 RCTs, four non-randomized trials, 24 single-arm trials, and four single-case designs. Random-effects meta-analyses estimated standardized mean differences for between-group and within-subject outcomes at post-treatment and follow-up. Risk of bias was formally assessed, and GRADE assessed evidence certainty.

Results: CBT produced moderate effects on global functioning at post-treatment (SMD = 0.47) that increased at follow-up (SMD = 0.67). QoL effects were smaller and non-significant at follow-up. Strongest effects were found for occupational functioning, effects on social functioning were moderate, and academic benefits were limited. Individual format and longer duration were associated with better outcomes. ADHD symptom improvement predicted QoL but not functioning.

Conclusions: CBT was associated with improvements in functioning (often the primary treatment concern), with moderate effects observed at post-treatment and some evidence of sustained effects at follow-up, particularly in directly targeted domains. The dissociation between symptom change, QoL, and functioning suggests that functional gains may not be fully explained by symptom reduction alone, although the mechanisms underlying these effects remain unclear. These findings support a more process-oriented perspective on CBT, while highlighting the potential relevance of directly targeting everyday functioning within psychological interventions. Risk of bias, self-report reliance, Western sample concentration, and short follow-ups (typically 3 months) warrant cautious interpretation.

1. Introduction

Attention deficit/hyperactivity disorder (ADHD) is a neurodevelopmental disorder characterized by a persistent pattern of inattention and/or hyperactivity-impulsivity that interferes with functioning (American Psychiatric Association, 2022). It often persists into adulthood, affecting approximately 3% of adults (Popit et al., 2024). Its main clinical characteristics are symptoms of inattention, hyperactivity,

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and/or impulsivity that cause functional impairment in important areas of life, reflecting disruptions in the psychological processes that regulate goal-directed behavior. Beyond these core symptoms, emotional dysregulation is increasingly recognized as a central feature of adult ADHD and one of the symptoms that causes the most functional impairment (Beheshti et al., 2020). Moreover, it often co-occurs with other psychiatric disorders, including anxiety, depression, and substance use disorders, among others (Choi et al., 2022). Additionally, it is significantly associated with a wide range of somatic conditions, including obesity, sleep disorders, asthma, and metabolic diseases (Libutzki et al., 2024).

1.1. Functional impairment and quality of life in adults diagnosed with ADHD

This clinical presentation is characterized not only by a broad range of core symptoms but also by marked impairment in everyday functioning (Cherkasova et al., 2022; Kosheleff et al., 2023). In fact, this impairment is often the main reason for consultation in adults rather than the ADHD symptoms *per se* (CADDRA, 2020). These symptoms also affect specific domains, including educational (DuPaul et al., 2021), occupational (Holst & Thorell, 2020), and social/familial (French et al., 2024) functioning, among others. In the academic domain, ADHD in adulthood has been linked with lower grade point averages (GPAs, Gormley et al., 2019), less frequent use of study skills (DuPaul et al., 2021), a lower academic level (Holst & Thorell, 2020), and a higher risk of educational dropout (Uchida et al., 2018). Regarding occupational functioning, adults diagnosed with ADHD are at an increased risk of having lower work performance (French et al., 2024), unemployment (Uchida et al., 2018), being fired (Fuermaier et al., 2021), and more frequent job changes (Helgesson et al., 2023). Regarding social and familial functioning, adult ADHD has been linked to smaller social networks, with fewer family members and friends (Holst & Thorell, 2020; Michielsen et al., 2015), as well as greater feelings of loneliness (Kosheleff et al., 2023), lower emotional connectedness (Montejo et al., 2019), and difficulties in establishing and maintaining friendships and achieving intimacy with others (French et al., 2024). Another relevant finding is that social network size has been shown to moderate the relationship between ADHD symptoms and functional impairment, with larger networks providing a protective effect (Reeble et al., 2023). Regarding sex differences, emerging evidence suggests that men diagnosed with ADHD may experience greater impairment in educational functioning, while women may experience greater impairment in social functioning (Faheem et al., 2022). Additionally, stronger predictors of functional impairment in adulthood include the persistence of ADHD symptoms from childhood and the presence of comorbidities (Cherkasova et al., 2022).

The quality of life (QoL) is significantly and negatively affected by symptoms and diagnosis of ADHD in adulthood, even when controlling for other demographic variables (French et al., 2024; Orm et al., 2023). This impact persists over time (Agnew-Blais et al., 2018). QoL captures an individual's perceived subjective well-being in their context (World Health Organization, 1995). In contrast, functional impairment refers to more objective limitations in life domains such as education, work, or social relationships. Reduced QoL applies not only to adults with a formal ADHD diagnosis, but also to those presenting with subclinical ADHD symptoms (Adamis et al., 2024). All dimensions of QoL (e.g., health-related, physical, emotional, and social QoL) are affected in adults with ADHD (Krauss & Schellenberg, 2022). It seems that higher comorbidities increase the risk of having a reduced QoL (Quintero et al., 2019). Furthermore, the QoL of friends and family members of individuals with ADHD is decreased as well (Zeides Taubin et al., 2024). Overall, the need for effective evidence-based treatments and rigorous evaluation of their impact on broader functional outcomes beyond the core symptoms is underscored by these functional impairments and reduced QoL in adults with ADHD.

1.2. Treatment of QoL and functioning in adults with ADHD

Both pharmacotherapy—primarily stimulant medication—and psychotherapy, particularly cognitive behavioral therapy (CBT), have been shown to improve core symptoms and comorbidities in adults with ADHD (Ostinelli et al., 2025). However, despite their clinical relevance, functional and QoL outcomes are often understudied, underreported, and overshadowed by symptom outcomes in ADHD clinical trials (Young et al., 2017). Since adults with ADHD present more negative automatic thoughts (Knouse et al., 2019) and greater behavioral avoidance (Torrente et al., 2014), CBT targets maladaptive cognitions and provides behavioral compensatory skills to address executive dysfunction (Ramsay, 2010), focusing on domain-specific functional impairment rather than core symptoms reduction. Additionally, third-wave behavioral approaches (e.g., acceptance and commitment therapy, mindfulness, and dialectical behavior therapy [DBT]) complement and broaden goals beyond symptom reduction toward valued-focused behaviors (Champ et al., 2021), which may further improve QoL and functioning.

Despite this theoretical rationale, the evidence base for CBT's impact on functional outcomes and QoL remains limited and methodologically heterogeneous. Coghill et al. (2017) found that effect sizes (ESs) were larger for stimulants than non-stimulants and for core symptoms than for QoL and functioning. Several meta-analytic reviews have explored CBT effects on functional outcomes. Knouse et al. (2017) found moderate post-treatment effects compared to controls on global functioning, which increased to moderate-to-large at follow-up according to López-Pinar et al. (2018). Other meta-analyses reported improvements in specific functioning domains, such as occupational (Lauder et al., 2024) and social functioning, even when delivered online (Shou et al., 2022). Moreover, combined CBT and pharmacotherapy may be more effective than pharmacotherapy alone for improving social functioning (Li & Zhang, 2024). Regarding QoL, Liu et al. (2023) found small-to-moderate effects at post-treatment, and López-Pinar et al. (2020) reported that these effects were sustained at follow-up. Zhang et al. (2025) reported similar findings for variants such as DBT.

However, previous reviews present several limitations, such as the inclusion of heterogeneous psychosocial interventions beyond CBT, focused exclusively on specific variants such as DBT, or examined only comparisons with pharmacotherapy. Notably, none comprehensively examined CBT effects on both QoL and functional impairment across specific functioning domains, limiting the clinical picture available to inform practice. Furthermore, given the substantial growth of the evidence base in recent years, previous reviews may have omitted a significant number of more recent studies. Overall, while research has largely focused on symptom improvement, existing findings suggest that CBT may benefit this population by improving functioning and QoL. Nevertheless, symptom reduction provides only an indirect indicator of change in everyday behavioral functioning, underscoring the need for a comprehensive synthesis focused on functional outcomes.

1.3. The present study

Functional impairment and QoL are central to the everyday challenges faced by adults with ADHD, yet research and previous reviews have largely focused on core and comorbid symptoms, often overlooking these broader outcomes. As a result, the field lacks a comprehensive understanding of how CBT influences real-world functioning, the mechanisms through which these changes may occur, and which treatment characteristics optimize functional outcomes. Addressing this gap requires moving beyond symptom-based evaluations toward a function-centered perspective that can inform both clinical decision-making and theoretical models of impairment. Accordingly, examining functional outcomes offers a unique opportunity to clarify how psychological interventions translate into real-world behavioral change in adult ADHD.

The present study addresses this gap by: (1) quantitatively

synthesizing the existing literature on the efficacy of CBT interventions in improving functional impairment and QoL in adults with ADHD; (2) examining the effects on specific domains of functioning (social, occupational and educational); (3) exploring how sample and study characteristics moderate these effects and may reveal underlying treatment processes; and (4) examining whether improvements in core ADHD symptoms predict changes in functional outcomes.

2. Method

This review was pre-registered in PROSPERO (CRD42024590094) and conducted in accordance with the PRISMA guidelines (Page et al., 2021). No funding was received from public, commercial, or nonprofit sources. The PRISMA checklist, the full screening and selection process, the codebook, and the extracted data can be found in the Open Science Framework (OSF) associated link.

2.1. Eligibility criteria

Studies were included if they met the following criteria:

- **Participants:** Adults (≥18 years) formally diagnosed with ADHD according to DSM-5 or ICD-11 criteria, by the research team or a community clinic, or reporting clinically elevated ADHD symptoms on validated ADHD symptom measures. Both formally diagnosed participants and those with clinically elevated ADHD symptoms were included to maximize the comprehensiveness of the review and capture the full breadth of available evidence in the field; subgroup analyses comparing these two groups were conducted to examine whether diagnosis status moderated treatment effects. Comorbidities (e.g., substance use disorder, depression, anxiety disorders, high-functioning autism) were allowed.
- **Intervention**: Eligible interventions included any structured psychological treatment grounded in behavioral or cognitive-behavioral principles, encompassing second-wave CBT (e.g., cognitive restructuring, metacognitive therapy), third-wave approaches (e.g., mindfulness-based cognitive therapy, dialectical behavior therapy), and psychoeducation programs explicitly aimed at developing cognitive or behavioral skills. All eligible interventions were required to explicitly introduce at least one specific cognitive or behavioral skill and to be adapted to the needs of adults with ADHD. These intervention types were subsumed under the CBT umbrella given their shared theoretical foundations in behavioral and cognitive principles (Hofmann & Asmundson, 2008). The intervention could be delivered alone or in combination with pharmacotherapy.
- **Control:** For RCTs, at least one arm did not receive a CBT-based intervention and therefore served as a control group. Controls could be active (e.g., other psychological treatment, medication alone, support groups, treatment as usual) or inactive (e.g., waitlist, no intervention). RCTs in which all arms received CBT were also eligible for within-subject analyses.
- **Outcomes**: Studies reported a measure of functional impairment and/or QoL, assessed by self-report (e.g., Sheehan Disability Scale, AAQoL) or by clinician/researcher rating.
- **Study design:** To provide a comprehensive synthesis of the literature, eligible designs included randomized controlled trials, single-arm trials, and single-case designs. Between-group and within-subject analyses were conducted and reported separately to distinguish controlled from uncontrolled treatment effects. Only articles published in peer-reviewed journals in English, German, Swedish, Dutch, or Spanish were considered. Language restrictions were applied based on the languages natively spoken by or proficient to the research team.

2.2. Information sources and search strategy

We conducted the primary search in MedLine (via PubMed), Scopus, and PsycInfo, without date restrictions. The last search was conducted on October 4, 2025. The search combined terms for population (ADHD, attention deficit) and intervention (CBT, cognitive behavioral therapy, dialectical behavior therapy). The full search strings are available at the OSF link. In addition, we performed backward and forward citation searches of all included studies (in Scopus) and screened the reference lists of relevant prior reviews.

2.3. Selection process

The first author conducted the initial screening of titles, abstracts, and keywords. Full-text screening was subsequently performed by the second and third authors. Disagreements were resolved by consensus. Interrater agreement was substantial ($\kappa = 0.75$). Screening was conducted using an online form, with reviewers not blinded to study characteristics.

2.4. Data collection and data items

Data extraction was carried out using a pre-defined template and codebook. The second and third authors independently extracted and coded the following variables: (A) number of study arms and arms receiving CBT; (B) country(es) in which the study was conducted; (C) number of participants at baseline; (D) dropouts in the CBT group(s); (E) sample characteristics (diagnosis, percentage of participants with at least one comorbidity, mean age, and percentage of males); (F) intervention characteristics (psychotherapy type [second-wave CBT, DBT, mindfulness, psychoeducation, or others], number of sessions or weeks, provider, delivery and format, type of medication, percentage of participants on ADHD medication); and (G) type of comparison group. Disagreements were resolved by consensus.

The first author extracted the data required for computing effect sizes (number of participants, pre-treatment standard deviation, and pre-, post-, and follow-up means where available). We contacted the corresponding authors for missing data. To ensure accuracy, a randomly selected 33% of these data points were independently verified by the fourth author.

2.5. Assessment of risk of bias

Risk of bias was assessed using study-design-specific tools: the Cochrane Risk of Bias Tool 2 for RCTs (Sterne et al., 2019), with the Metapsy adaptation for psychotherapy trials (Miguel et al., 2025), ROBINS-I for nonrandomized controlled trials (Sterne et al., 2016), and the NIH Tool for before-after studies with no control group for single-arm trials (National Institutes of Health, 2021). Risk of bias assessment was conducted independently by the first and fourth authors, with disagreements resolved by consensus. Publication bias was examined through visual inspection of funnel plot symmetry, the trim-and-fill method (Duval & Tweedie, 2000), and Egger's test (Egger et al., 1997).

2.6. Assessment of certainty of evidence

The certainty of evidence for each outcome was assessed using the GRADE (Grading of Recommendations Assessment, Development and Evaluation) framework (Guyatt et al., 2008), evaluating risk of bias, inconsistency, indirectness, imprecision, and publication bias to rate evidence as high, moderate, low, or very low certainty. GRADE assessments were conducted by the first author and reviewed by the fourth author, with disagreements resolved by consensus.

2.7. Data analysis

Post-treatment and follow-up changes in QoL and global functional impairment were defined as the primary outcomes. Domain-specific functioning (academic, occupational, and social) was examined as a secondary outcome.

For the meta-analysis, standardized mean differences (SMDs) and their variances were calculated using Morris's formulas for within-subject (Morris, 2000) and between-group outcomes (Morris, 2008), which include a small sample factor correction (c_p):

$$d = c_p \frac{\left[\left(M_{postT} - M_{preT}\right) - \left(M_{postC} - M_{preC}\right) \right]}{SD_{prePooled}}$$

For estimating the variances of both effect size indices, as the pre-post correlation was not reported in the studies, a value of 0.70 was assumed for r, as recommended by Rosenthal (1991). Sensitivity analyses with r values of 0.50 and 0.90 were also conducted. Standard errors were derived from the square root of the estimated variance for each SMD. Following Cohen (1988), SMDs of 0.2, 0.5, and 0.8 were interpreted as small, moderate, and large effects, respectively. Given the expectation of high heterogeneity and the assumption that true effects could vary across studies (Borenstein et al., 2010), random-effects models were used in all meta-analyses, with a 95% confidence interval. Heterogeneity was quantified with the I^2 index, following standardized interpretation, i.e., 25% = small, 50% = moderate, and >75% high heterogeneity, and with Cochrane's Q (Higgins et al., 2003). A meta-analysis of the included SCDs was attempted, but no study included enough measures (<3 observations) to compute the ES or the associated standard error. Therefore, a narrative review approach was adopted for these studies.

2.7.1. Additional analyses

Potential moderators were examined, including sample characteristics (dropouts, diagnosis, proportion with comorbidities, mean age, and proportion of males), intervention features (type of psychotherapy, number of sessions or weeks, provider, delivery format, type of medication, proportion of participants on ADHD medication), and control group type. Subgroup analyses were conducted for categorical moderators, and meta-regressions for continuous moderators. They were restricted to subgroups with >5 studies and meta-regressions to outcomes with >10 studies, as fewer studies yield unstable estimates and increase the risk of spurious findings (Borenstein et al., 2009). We conducted sensitivity analyses using a leave-one-out approach to evaluate the influence of individual studies on the pooled effects. Additional sensitivity analyses excluding studies with high risk of bias were conducted for all outcomes. Studies were classified as high risk if rated as high risk of bias (RoB 2), critical/serious risk (ROBINS-I), or poor quality (NIH tool). To evaluate statistical power, we calculated minimum detectable effects (MDEs) for each meta-analysis, defined as the smallest standardized mean difference detectable with 80% power at a two-tailed α of 0.05, given the observed standard errors.

All the analyses were performed in R (R Core Team, 2025) using the *meta* package (Balduzzi et al., 2019). The full R code and database are available in OSF at this link.

3. Results

Of the 2484 records reviewed, 70 studies were included (Table 1, Fig. 1): 38 randomized controlled trials (RCTs), four nonrandomized controlled trials, 24 single-arm trials, and four SCDs with 5152 participants. Full references for included studies are shown in Supplementary Materials. Most of the controlled trials (69.1%) used an active comparator (e.g., standard clinical management, parallel loosely structured discussion group, treatment as usual, among others). Over half of the studies (55%) were conducted in Europe, with most of those (33%) in Sweden. Thirty percent were conducted in the Americas, primarily in the United States (90%), and 15% were conducted in Asia, with half of those in China. Participants were, on average, 32.6 years old, and 47.8% were male. Of these participants, 84.5% were formally diagnosed with ADHD, and 57.4% had at least one comorbidity. Interventions were primarily stand-alone second-wave CBT (53.5%), followed by third-wave approaches, such as DBT or mindfulness (8.4% each). These interventions were typically delivered by therapists in face-to-face (74.6%) vs remote (15.7%), and individual (44%) vs group (34.2%) or combined (9.8%) formats across an average of 12 sessions. The mean dropout rate was 14.9%, and 62.1% of participants received ADHD medication.

3.1. Risk of bias assessment results

Of the RCTs, 36% were assessed as having high risk of bias, 31% low risk, and 33% some concerns (Fig. 2). All non-randomized trials (100%) had serious or critical risk of bias. Among single-arm trials, 46% had good methodological quality, 46% fair, and 8% poor quality. One of the most frequent issues among the RCTs was the insufficient reporting of allocation concealment. The main concerns for nonrandomized controlled trials were inadequate control of confounding variables, selection bias in assigning participants to groups, and deviations from intended interventions. Common issues among single-arm trials included non-representative samples (e.g., high-functioning participants and unbalanced sex ratios) and small sample sizes. Across all study designs, issues such as inappropriate handling of missing data and lack of pre-registration could introduce bias in the findings.

Publication bias analyses (see Supplementary Table 1) revealed potential bias in the following outcomes only: between-group QoL, academic functioning, and social functioning at follow-up; and within-subject global functioning. This is suggested by significant Egger's tests and trim-and-fill adjustments that rendered the effects nonsignificant. On the other hand, visual inspection of the funnel plots did not reveal asymmetry clearly. For all other outcomes, the tests were non-significant, suggesting the absence of publication bias.

3.2. Meta-analyses results

Full results are in Table 2; forest plots for primary outcomes in Figs. 3–4; remaining forest plots and funnel plots in Supplementary Figs. 1–29. The pooled estimates reported below should be interpreted in light of the risk of bias assessment results, which indicate that a substantial proportion of included studies carried methodological limitations.

Global functioning improved with moderate effects that increased over time (k = 16, n = 1,164, SMD = 0.47, 95% CI [0.32, 0.62] at post-treatment; k = 8, n = 703, SMD = 0.67, 95% CI [0.52, 0.82] at follow-up). QoL effects were smaller: small-to-moderate improvements at post-treatment (k = 25, n = 2,249, SMD = 0.28, 95% CI [0.14, 0.42]) did not persist at follow-up (k = 10, n = 993, SMD = 0.18, 95% CI [−0.03, 0.40]). Heterogeneity was moderate (I^2 = 30.3-77.9%). Cochrane's Q was significant for most outcomes, especially the primary ones, confirming the presence of variability beyond chance. Additionally, forest plots revealed considerable spread in study-level effects, including several near-zero and negative effects, indicating that pooled estimates should not be interpreted as reflecting a uniform benefit across all studies. Across specific domains, occupational functioning showed the largest effects. The follow-up SMD was also large, although it was only based on two studies and must be considered preliminary. Social functioning improved with small-to-moderate effects that persisted at follow-up, whereas academic functioning showed short-term gains that were not maintained at follow-up.

Within-subject effects were moderate for global functioning and QoL. At follow-up they remained stable for QoL and increased to large for global functioning. Heterogeneity was high for within-subject

Table 1
Main characteristics of the included studies: Country in which the study was conducted; study design; number of arms in the study; number of arms receiving CBT; number of participants in the baseline; percentage of dropouts in CBT group/s; Dx = diagnosis of the participants; percentage of participants with at least one comorbidity; mean age of the participants; percentage of men within the sample; type of intervention; length of the intervention, in sessions or weeks; provider of the intervention (therapist vs self-delivered); delivery of the intervention; format of the intervention; type of medication and percentage of participants on medications for ADHD; for controlled studies, comparison/s.

Study	Country	Design	# of arms	# of CBT arms	Base-line *n*	Drop-outs (%)	Sample characteristics				Intervention/s characteristics							
							Dx	% Comorbidity	Mean Age	% Men	Intervention	Length (sessions/ weeks)	Applied by	Delivery	Format	Medication	% on meds	Comparison
Anastopoulos 2018	United States	Single-arm trial	1	1	88	9.1	ADHD	56	20.2	40.9	CBT	8	Therapist	Face-to-face	Combo	-	52.4	-
Bueno 2015	Brazil	Non-RCT	4	1	48	8.33	ADHD	-	29.62	50	Mindfulness	8	Therapist	Face-to-face	Group	Stimulants	70.8	-
Carroll 2023	Sweden	Single-arm trial	1	1	213	20	ADHD	74.7	34.89	24.7	CBT + DBT + ACT	14 + 7	Therapist	Face-to-face	Combo	-	73.7	-
Cherkasova 2016	Canada	RCT	2	2	88	13.87	ADHD	18.18	34.6	69.32	CBT	12 + 12	Therapist	Face-to-face	Combo	Stimulants	47.7	CBT vs CBT + meds
Corbisiero 2018	Switzerland	RCT	2	1	43	4.65	ADHD	37.21	32.03	55.81	CBT	12-14	Therapist	Face-to-face	Individual	Stimulants	100	Standard clinical management
Corrales 2024	Spain	RCT	2	2	81	-	ADHD	55.6	41.27	58	CBT	6 and 12	Therapist	Face-to-face	Group	Combo	100	6 session CBT vs 12 session CBT
Dittner 2018	United Kingdom	RCT	2	1	60	23.3	ADHD	-	35.9	68.33	CBT	16	Therapist	Face-to-face	Individual	Combo	75	Treatment as usual
Eddy 2015	United States	SCD	1	1	4	0	ADHD	0	21.75	75	CBT	8	Therapist	Face-to-face	Individual	Combo	50	-
Eddy 2021	United States	RCT	2	1	250	20.2	ADHD	60	19.68	44	CBT	9 + 8	Therapist	Face-to-face	Combo	Combo	47.2	Wait-list
Emilsson 2011	Iceland	RCT	2	1	54	26	ADHD	85.2	33.8	37.03	DBT	15	Therapist	Face-to-face	Combo	Combo	100	Treatment as usual
Eto 2025	Japan	RCT	2	1	30	0	ADHD	53.33	35.75	53.33	CBT	12	Therapist	Remote	Individual	Combo	100	Treatment as usual
Fleming 2015	United States	RCT	2	1	33	5.89	ADHD	-	21.35	57.57	DBT	9 + 7	Therapist	Combo	Combo	Combo	75.75	Skills handouts
Grinblat 2023	Israel	RCT	2	1	46	13.3	ADHD	-	35.65	30.4	Mindfulness	11	Therapist	Combo	Individual	-	-	Wait-list
Gu 2017	China	RCT	2	1	54	6.66	ADHD	-	20.29	55.5	Mindfulness	6	Therapist	Face-to-face	Individual	Combo	70.4	Wait-list
Gutman 2020	United States	RCT	2	1	25	8.33	No Dx	-	30.86	0	Occupational Therapy CBT	7	Therapist	Face-to-face	Individual	Combo	69.56	No treatment
Halmøy 2022	Norway	RCT	2	1	121	15	ADHD	66	37	43.8	DBT	14	Therapist	Face-to-face	Combo	Combo	63	Treatment as usual
Hartung 2022	United States	Single-arm trial	1	1	30	0	ADHD	70	22.6	57	CBT	9	Therapist	Face-to-face	Combo	-	-	-
Hepark 2014	Netherlands	Single-arm trial	1	1	11	18.18	ADHD	45.5	-	63.6	Mindfulness	10	Therapist	Face-to-face	Group	Combo	72.7	-
Hepark 2015	Netherlands	RCT	2	1	103	22.64	ADHD	-	35.85	55.3	Mindfulness	12	Therapist	Face-to-face	Group	Combo	57.28	Wait-list
Hiltunen 2014	Finland	RCT	2	1	17	10	ADHD	77.7	35.5	35.3	CBT	10	Therapist	Face-to-face	Individual	Stimulants	55.5	Hypnotherapy
Hirvikoski 2011	Sweden	RCT	2	1	51	19.2	ADHD	72.5	38.9	37.25	DBT	14	Therapist	Face-to-face	Group	Combo	56.86	Parallel loosely structured discussion group
Hirvikoski 2017	Sweden	RCT	2	1	87	4.2	ADHD	64.6	38.6	41.3	PsyEd	8	Therapist	Face-to-face	Group	Combo	62.06	Treatment as usual

(*continued on next page*)

Table 1 (*continued*)

Study	Country	Design	# of arms	# of CBT arms	Baseline *n*	Drop-outs (%)	Sample characteristics				Intervention/s characteristics							Comparison
							Dx	% Comorbidity	Mean Age	% Men	Intervention	Length (sessions/ weeks)	Applied by	Delivery	Format	Medication	% on meds	
Hoxhaj 2018	Germany	RCT	2	2	81	7	ADHD	15.87	39.5	48.14	Mindfulness/ PsyEd	8	Therapist	Face-to-face	Group	Combo	7.4	Mindfulness vs psyEd
Huang 2019	China	RCT	3	2	108	23.3	ADHD	38.88	26.66	62.5	CBT	12/15 (booster)	Therapist	Face-to-face	Group	Combo	31.48	Wait-list
Janssen 2018	Netherlands	RCT	2	1	120	15	ADHD	58.33	39.35	47	Mindfulness	8	Therapist	Face-to-face	Group	-	54.16	Treatment as usual
Janssen 2020	Netherlands	Single-arm trial	1	1	31	16	ADHD	52	38	48	Mindfulness	8	Therapist	Face-to-face	Group	-	55	-
Jensen 2021	Norway	Single-arm trial	1	1	32	37.5	ADHD	87.5	35.7	56.25	Goal management training	9	Therapist	Face-to-face	Group	-	50	-
Kenter 2023	Norway	RCT	2	1	120	28.3	No Dx	-	40.9	20	CBT + DBT + Goal Management training	7	Self-applied	Remote	Individual	-	83.3	Online psychoeducation module
Knouse 2022	United States	Single-arm trial	1	1	240	61.25	No Dx	-	29.15	21.3	CBT	7	Self-applied	Remote	Individual	-	60.4	-
LaCount 2015a	United States	Single-arm trial	1	1	17	29.41	No Dx	-	25.41	35.29	CBT	12	Therapist	Face-to-face	Combo	Stimulants	64.7	-
LaCount 2015b	United States	Non-RCT	1	1	41	12	No Dx	-	21.4	64.9	CBT	3	Therapist	Face-to-face	Group	-	5.4	No intervention
Langer 2013	Germany	Single-arm trial	1	1	18	5.55	ADHD	-	34.4	38.88	Stress management training	4	Therapist	Face-to-face	Group	Stimulants	44.4	-
López-Pinar 2023	Spain	SCD	1	1	2	0	ADHD	50	21	0	CBT	11	Therapist	Remote	Individual	Stimulants	100	-
Lücke 2021	Germany	RCT	4	2	433	3.24	ADHD	-	35	50.1	DBT	22	Therapist	Face-to-face	Group	Stimulants	50	Clinical management
Meinzer 2021	United States	RCT	2	1	113	23.64	ADHD	-	19.87	49.1	Motivational interviewing + behavioral activation	5	Therapist	Face-to-face	Individual	-	68.14	Supportive counseling
Moëll 2015	Sweden	RCT	2	1	57	10.34	No Dx	36.84	36.8	31.58	CBT	6	Combo	Remote	Individual	-	-	Wait-list
Morgensterns 2016	Sweden	Single-arm trial	1	1	98	29.41	ADHD	71.1	37.4	31.6	DBT	14	Therapist	Face-to-face	Group	Combo	74.7	-
Nakashima 2021	Japan	RCT	2	1	48	4.2	ADHD	-	39.11	12.5	CBT	8	Therapist	Face-to-face	Individual	Combo	66.77	Treatment as usual
Nasri 2020	Sweden	Single-arm trial	1	1	18	0	ADHD	66	39.7	66	CBT + DBT	14	Therapist	Face-to-face	Group	-	72	-
Nasri 2023	Sweden	RCT	3	1	104	11.1	ADHD	-	36.63	30.77	CBT + DBT	12	Self-applied	Remote	Individual	Combo	67	Treatment as usual
Nordby 2021	Sweden	Single-arm trial	1	1	13	46	ADHD	-	39.2	46.2	CBT + DBT	3	Self-applied	Remote	Individual	-	84.6	-
Nordby 2024	Sweden	Single-arm trial	1	1	16	37.5	ADHD	-	-	37.5	DBT + positive psychology	8	Combo	Combo	Combo	-	81.3	-

(*continued on next page*)

Table 1 (*continued*)

Study	Country	Design	# of arms	# of CBT arms	Baseline *n*	Drop-outs (%)	Sample characteristics				Intervention/s characteristics							Comparison
							Dx	% Comorbidity	Mean Age	% Men	Intervention	Length (sessions/ weeks)	Applied by	Delivery	Format	Medication	% on meds	
Oguchi 2025	Japan	Single-arm trial	1	1	29	10.3	No Dx	0	20.76	58.6	CBT	6	Therapist	Face-to-face	Individual	-	-	-
Oscarsson 2025	Sweden	Single-arm trial	1	1	36	-	ADHD	-	44.4	19	CBT	12	Self-applied	Remote	Individual	-	86	-
Pagan 2025	United States	Single-arm trial	1	1	9	0	ADHD	100	35.63	30	CBT	13.5	Therapist	Face-to-face	Individual	-	40	-
Pan 2019	China	Non-RCT	2	2	124	18.54	ADHD	49.19	26.96	54.03	CBT	12	Therapist	Face-to-face	Group	Combo	45.96	CBT vs CBT + meds
Pan 2022	China	RCT	2	1	98	10.2	ADHD	44.89	25.81	59.18	CBT	12	Therapist	Face-to-face	Group	Combo	100	Treatment as usual
Petterson 2017	Sweden	RCT	3	2	45	CBTg: 7.14, iCBT: 15.38	ADHD	-	37.45	35.55	CBT	10	Combo/ Self-applied	Combo/ Remote	Combo	-	48.9	Wait-list
Puente 2016	United States	SCD	1	1	1	0	ADHD	0	32	100	CBT	12	Therapist	Face-to-face	Individual	No	0	-
Ramsay 2011	United States	Single-arm trial	1	1	5	0	ADHD	40	36.6	80	CBT	16	Therapist	Face-to-face	Individual	No	0	-
Rostain 2006	United States	Single-arm trial	1	1	74	13.51	ADHD	83.7	30.8	74.4	CBT	16	Therapist	Face-to-face	Individual	Stimulants	81.2	-
Schoenberg 2014	Netherlands	RCT	2	1	61	18.75	ADHD	-	36.7	47.73	Mindfulness	12	Therapist	Face-to-face	Group	Stimulants	54	Wait-list
Selaskowski 2022	Germany	RCT	2	2	60	28.3	ADHD	-	38.2	53.49	PsyEd	8	Combo	Combo	Group	Combo	48.84	App supported group PsyEd vs pen-and-paper brochures
Selaskowski 2023	Germany	RCT	2	2	40	15	ADHD	-	29.6	47.06	PsyEd	-	Self-applied	Remote	Individual	Combo	55.88	Self-guided PsyEd based on a chatbot or an App
Solanto 2020	United States	Single-arm trial	1	1	19	5.26	ADHD	44	23.61	44	CBT	12	Therapist	Face-to-face	Group	Stimulants	28	-
Stralin 2022	Sweden	Single-arm trial	1	1	39	33.33	ADHD	-	31.3	46.1	CBT	14	Therapist	Face-to-face	Group	-	-	-
Stralin 2025	Sweden	RCT	2	2	108	-	ADHD	-	38.9	36.4	CBT/DBT	14	Therapist	Face-to-face	Group	-	32.4	CBT vs DBT
Tsirmpas 2023	Greece	Single-arm trial	1	1	30	3	No Dx	100	31.44	33	CBT + mindfulness + positive psychology	16	Combo	Remote	Individual	-	-	-
Ulusoy 2024	Turkey	RCT	2	1	32	37.5	ADHD	62.5	29.16	37.5	DBT	12	Therapist	Remote	Group	-	62.5	Wait-list
van Emmerik 2015	Netherlands	SCD	1	1	2	50	ADHD + SUD	100	28	100	CBT	15	Therapist	Face-to-face	Individual	-	-	-
van Emmerik 2019	Netherlands	RCT	2	1	119	22	ADHD + SUD	100	35.1	83.2	CBT	15	Therapist	Face-to-face	Individual	Stimulants	4.2	Standard CBT for SUD
Vidal 2013	Spain	RCT	2	1	32	6.25	ADHD	-	39.47	53.10	CBT	12	Therapist	Face-to-face	Group	Combo	100	PsyEd not focused on skills building

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Table 1 (*continued*)

Study	Country	Design	# of arms	# of CBT arms	Baseline *n*	Drop-outs (%)	Sample characteristics			% Men	Intervention/s characteristics							Comparison
							Dx	% Comorbidity	Mean Age		Intervention	Length (sessions/ weeks)	Applied by	Delivery	Format	Medication	% on meds	
Virta 2010	Finland	RCT	3	1	29	-	ADHD	48.27	34.73	48.27	CBT	10	Therapist	Face-to-face	Individual	Stimulants	58.6	Cognitive training and wait-list
Weiss 2012	United States	RCT	2	2	48	10.42	ADHD	-	35.6	72.60	CBT	9	Therapist	Face-to-face	Individual	Stimulants	47.9	CBT + meds vs CBT + placebo
Wettstein 2021	Netherlands	Non-RCT	2	1	627	13	ADHD	-	29.75	52.00	CBT	11.5	Therapist	Face-to-face	Individual	Combo	100	Treatment as usual
Wilens 1999	United States	Single-arm trial	1	1	26	0	ADHD	96	42.4	85.00	CBT	36	Therapist	Face-to-face	Individual	Stimulants	58	-
Wymbs 2015	United States	Single-arm trial	1	1	15	0	ADHD	-	29.4	50.00	CBT	6	Therapist	Face-to-face	Group	Stimulants	40	-
Young 2015/17	United Kingdom	RCT	2	1	95	27	ADHD	-	35.17	43.70	CBT	15	Therapist	Face-to-face	Combo	Combo	100	Treatment as usual
Yovel 2007	United States	Single-arm trial	1	1	16	0	ADHD	-	42.67	8.00	CBT	12-15	Therapist	Face-to-face	Individual	-	100	-
Zhang 2025	China	RCT	2	1	86	0	ADHD	-	25	47.70	CBT	1-12	Therapist	Remote	Group	Combo	100	Treatment as usual

Note. - = data not available; RCT = Randomized Controlled Trial; Non-RCT = Non-Randomized Controlled Trial; SCD = Single-Case Design; SUD = substance use disorder; DBT = Dialectical Behavior Therapy; No dx = No formal ADHD diagnosis (self-reported or probable ADHD); Combo = Combined.

outcomes (I^2 = 71.2-88.6%). For the specific functioning domains, the pattern was similar to that observed for between-group effects.

Leave-one-out sensitivity analyses showed that results were robust across all outcomes. All results were robust to sensitivity analyses using r values of 0.50 and 0.90. Sensitivity analyses excluding high risk of bias studies revealed no significant differences in effect sizes between high and non-high risk studies. Effects remained statistically significant when analyzing only non-high risk studies, except for between-group academic functioning. Achieved statistical power was generally high except for QoL and academic functioning at follow-up. GRADE certainty for between-group outcomes was high for global functioning and moderate for most other outcomes, but low to very low for academic and occupational functioning at follow-up due to small sample sizes. For within-subject outcomes, certainty was high for social functioning at post-treatment, moderate for most primary and secondary outcomes, and low to very low for outcomes with limited evidence or concerns about publication bias.

3.3. Subgroup analyses results

Statistically significant subgroup analyses are reported in Supplementary Table 2. Few significant moderator effects emerged. Individual format (k = 7) was more effective than group format (k = 7) for global functioning in both between-group and within-subject analyses. For within-subject global functioning, second-wave CBT (k = 24) was significantly more effective than mindfulness (k = 5). Additionally, remote delivery (k = 7) outperformed face-to-face (k = 16) interventions for between-group QoL, although this finding should be interpreted with caution given the small number of studies in each subgroup and the potential for unexamined differences between them. All other subgroup comparisons were nonsignificant or could not be conducted due to insufficient studies (k < 5), including study design (RCT vs. non-RCT), diagnostic status (formal diagnosis vs. high symptomatology), provider (therapist vs. self-delivered), medication type (stimulant vs. combination), or comparator type (active vs. inactive control).

3.4. Meta-regression analysis results

Significant meta-regressions are shown in Supplementary Table 3. For QoL, two consistent predictors emerged. Higher psychiatric comorbidity rates were associated with poorer within-subject outcomes at post-treatment (R^2 = 13.7%, k = 27). Greater ADHD symptom reductions predicted better QoL outcomes between-group (R^2 = 29.0%, k = 23) and within-subject (R^2 = 22.7%, k = 17) at post-treatment, and at follow-up (R^2 = 68.1%, k = 11; Fig. 5).

For global functioning, several study characteristics were significant. Higher male proportions predicted greater within-subject improvements at post-treatment (R^2 = 19.2%, k = 36) and follow-up (R^2 = 47.7%, k = 10). Longer treatment duration predicted larger within-subject post-treatment effects (R^2 = 28.7%, k = 36). Additionally, greater ADHD symptom reductions predicted stronger post-treatment improvements (R^2 = 20.4%, k = 35; Fig. 6).

For specific domains, findings were limited. Older age predicted larger within-subject academic functioning effects (R^2 = 87.4%, k = 12), while higher medication rates predicted larger between-group social functioning effects (R^2 = 89.2%, k = 13). However, these very high R^2 values should be interpreted cautiously given the small number of studies and potential overfitting. More generally, all meta-regression findings should be interpreted with caution, as high R^2 values in the context of a limited number of studies may reflect instability or leverage from individual trials rather than robust moderating effects. All other meta-regressions were non-significant.

3.5. SCDs narrative review

Across the SCDs, CBT was generally linked to improvements in

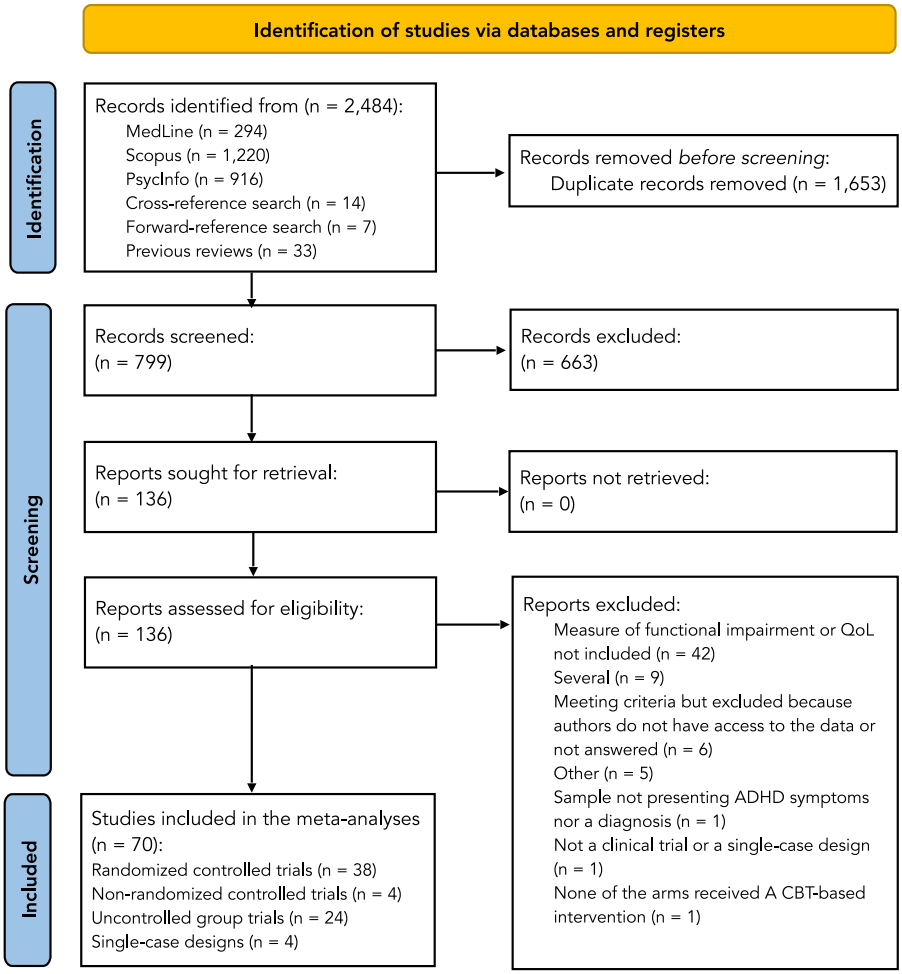


Fig. 1. PRISMA flow diagram.

functioning and QoL, though durability varied. In college populations, both face-to-face (Eddy et al., 2015) and online delivery (López-Pinar & Vicente-Gispert, 2023) reduced impairment and improved organization, time management, and social/academic engagement (e.g., better grades, more time with friends), with the online cases showing QoL gains that increased up to 6-month follow-up. A community SCD (Puente & Mitchell, 2016) also found enhanced functioning and QoL post-treatment, but partial relapse and interpersonal difficulties at 2-month follow-up. Finally, in cases with co-occurring substance use disorder (van Emmerik-van Oortmerssen et al., 2015), CBT improved ADHD symptoms, substance use, and QoL, although sustained benefits were observed only in the participant who remained engaged. Together, these findings suggest CBT can yield meaningful improvements in functioning and QoL across settings but highlight the need for strategies to maintain gains over time.

4. Discussion

Traditionally, literature on interventions for adults with ADHD has focused on improving core symptoms, with moderate effects for CBT (Liu et al., 2023) and stimulants (Gosling et al., 2025). This meta-analysis summarizes the evidence on how CBT effects in adult ADHD relate to changes in everyday behavioral functioning beyond symptom reduction. Specifically, we quantified the impact of CBT on global functioning and QoL, examined effects across specific functional domains (social, occupational, and educational), explored potential moderators of treatment effects, and assessed whether improvements in core ADHD symptoms predict functional outcomes.

Compared to previous studies, this review substantially expands the evidence base by including 70 studies with 5152 participants, surpassing the 32 studies with 896 participants in Knouse et al. (2017) and the 28 RCTs with 2190 participants in Liu et al. (2023). The continued growth of this literature may reflect the increasing recognition of CBT's potential for adults with ADHD. The characteristics of the included sample align with epidemiological data on adults with ADHD, showing comparable rates of comorbidity (Faraone & Larsson, 2019) and a higher proportion receiving medication (62.1%), which is typical of treatment-seeking samples included in clinical trials (Garcia-Argibay et al., 2025). The interventions predominantly represented the most commonly implemented model to date: face-to-face, second-wave CBT delivered over an average of 12 sessions. Average dropout rates (14.9%) were lower than those typically reported in CBT trials (Fernandez et al., 2015), which may suggest favorable treatment acceptability in this population. This expanded evidence base provides the most comprehensive assessment of CBT's effects on functioning and QoL in adults with ADHD to date.

4.1. CBT effect on QoL and global functioning

Across the different outcomes examined, CBT showed the most consistent effects on global functioning, with moderate effects at post-treatment that increased at follow-up, with GRADE ratings indicating moderate-to-high certainty of evidence. These results should nonetheless be interpreted in light of the methodological limitations identified across studies. These findings build upon and extend those of previous reviews (Knouse et al., 2017; López-Pinar et al., 2018). The observed effects may reflect clinically relevant improvements in functioning,

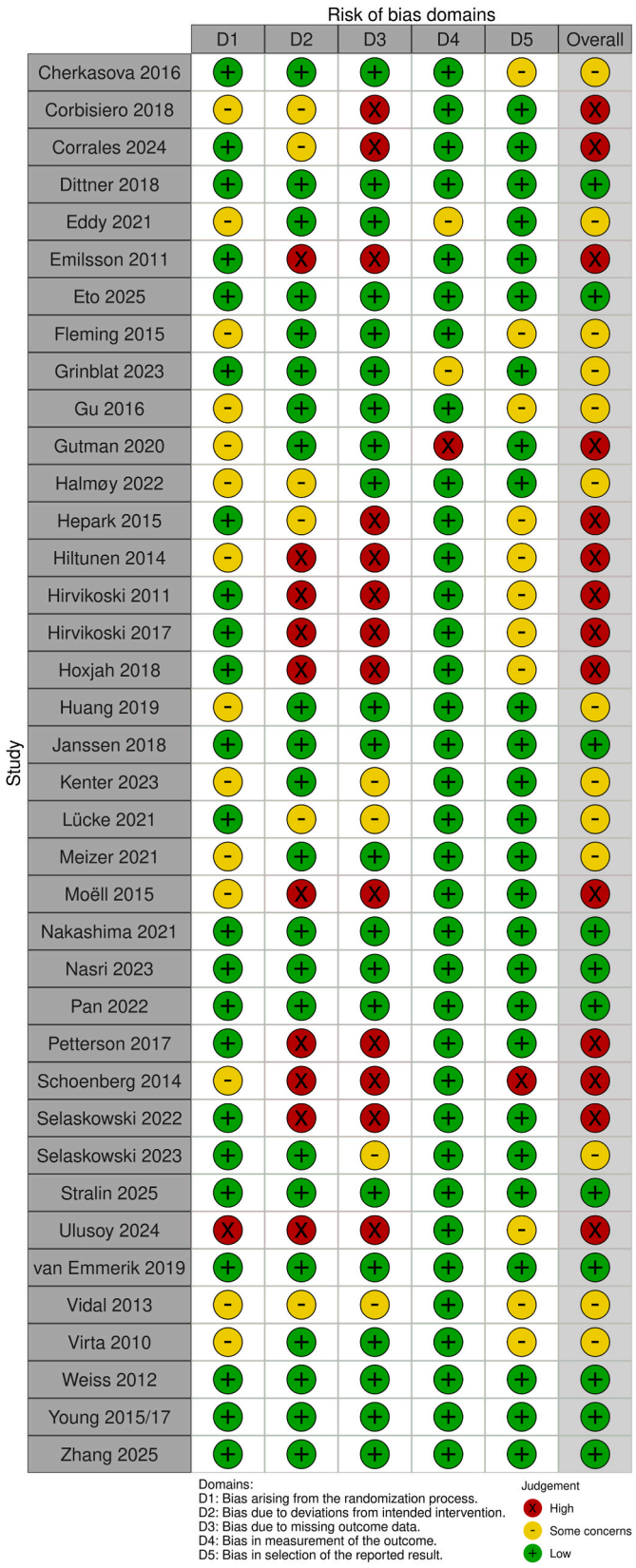
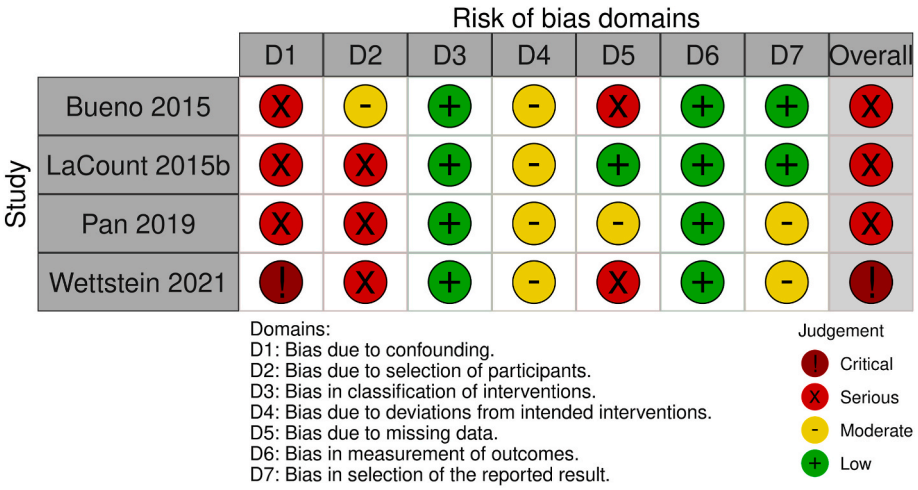
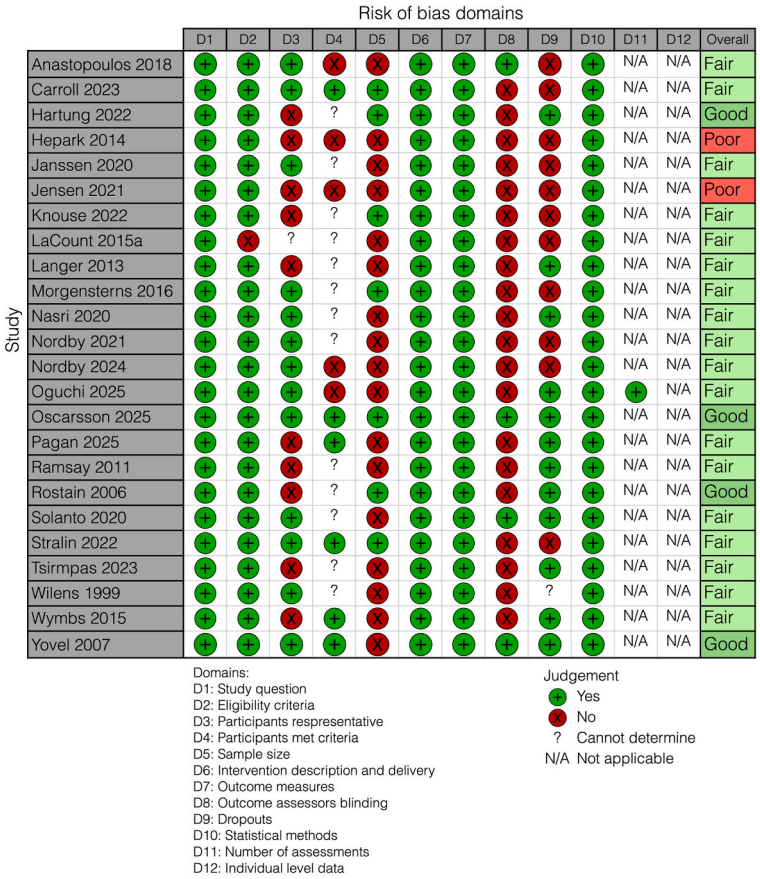


Fig. 2. Risk of bias summary.

which is noteworthy given that adults with ADHD typically seek treatment for functional impairment rather than symptom severity *per se* (Kosheleff et al., 2023). Given that ADHD symptoms typically persist across the lifespan (Cherkasova et al., 2022), CBT's focus on developing adaptive coping skills rather than eliminating symptoms may make it particularly suited for addressing the functional difficulties that



B



C

Fig. 2. (*continued*).

characterize adult ADHD (French et al., 2024). Our results are broadly consistent with previous literature suggesting that CBT may improve functional outcomes across disorders characterized by executive and emotional regulation deficits, including emotional disorders (Kambeitz-Ilankovic et al., 2022) and other neurodevelopmental disorders (Schweizer et al., 2023). Although the mechanisms underlying these effects were not directly tested in this meta-analysis and remain speculative, the consistency of functioning improvements across ADHD, emotional, and other neurodevelopmental disorders suggests that CBT's benefits may operate through transdiagnostic mechanisms, such as improved emotion regulation and cognitive flexibility, rather than disorder-specific pathways (Sloan et al., 2017), although this interpretation should be considered tentative.

In contrast, between-group effects on QoL were smaller and did not persist at follow-up. These results differ from López-Pinar et al. (2020), who found sustained follow-up effects on QoL, but align with Liu et al.'s (2023) post-treatment findings. This discrepancy may reflect differences in sample size and methodological characteristics across reviews,

Table 2
Standardized mean differences (SMD), 95% confidence intervals, heterogeneity analyses (I^2, Cochrane's Q), achieved statistical power and risk of bias for between-groups and within-group outcomes.

Variable	Assessment	Between-groups outcomes								Within-subject outcomes						
		k	n	SMD	95% CI	I^2	Q	GRADE	Power	k	n	SMD	95% CI	I^2	Q	GRADE	Power
QoL	PT	25	2249	0.28	0.14,0.42	73.8	91.77***	⊕⊕⊕○	97.2%	44	1908	0.58	0.46,0.70	88.6	377.87***	⊕⊕⊕○	100.0%
	FU	10	993	0.18	−0.03,0.40	77.9	45.23***	⊕⊕○○	31.8%	16	752	0.55	0.43,0.67	71.2	55.47***	⊕⊕⊕○	100.0%
Global functional impairment	PT	16	1164	0.47	0.32,0.62	64	41.71***	⊕⊕⊕⊕	95.2%	36	1246	0.58	0.45,0.70	80.3	177.96***	⊕⊕○○	100.0%
	FU	8	703	0.67	0.52,0.82	30.3	11.48	⊕⊕⊕⊕	66.7%	10	466	0.91	0.65,1.17	87.9	74.54***	⊕⊕⊕○	99.8%
Academic functioning	PT	8	460	0.48	0.17,0.79	61.5	18.16*	⊕⊕○○	58.9%	12	372	0.26	0.11,0.40	52.1	22.98*	⊕⊕⊕○	97.2%
	FU	3	155	0.04	−0.13,0.22	19	2.47	⊕○○○	7.2%	4	233	0.15	0.04,0.26	57.6	7.08	⊕⊕○○	77.9%
Occupational functioning	PT	8	220	1.00	0.62,1.38	58.8	16.99*	⊕⊕⊕○	82.1%	10	128	0.74	0.56,0.92	28	12.51	⊕⊕⊕○	100.0%
	FU	2	33	0.98	0.53,1.42	6.9	1.07	⊕○○○	83.0%	4	60	1.06	0.64,1.47	39	4.92	⊕⊕○○	99.9%
Social relationships functioning	PT	13	962	0.36	0.18,0.55	65.6	34.89***	⊕⊕⊕○	31.4%	24	859	0.33	0.26,0.40	31.4	33.51	⊕⊕⊕⊕	100.0%
	FU	5	526	0.37	0.10,0.65	72	17.84**	⊕⊕⊕○	97.2%	5	256	0.33	0.22,0.44	0	4.68	⊕⊕⊕○	100.0%

Note. PT = post-treatment assessment; FU = follow-up assessment; SMD = standardized mean differences; RoB = Risk of bias; * = $p < .05$; ** = $p < .01$; *** = $p < .001$; ⊕○○○ = very low certainty/quality of evidence; ⊕⊕○○ = low certainty/quality of evidence; ⊕⊕⊕○ = moderate certainty/quality of evidence; ⊕⊕⊕⊕ = high certainty/quality of evidence.

although this explanation should be interpreted cautiously. However, within-subject QoL effects were moderate and appeared stable over time, supporting previous studies (López-Pinar et al., 2020). These differential effects between the two primary outcomes may reflect several factors, though these explanations were not directly tested and should be considered speculative. First, the more specific and objective nature of functioning versus the more subjective well-being nature of QoL (World Health Organization, 1995). Second, temporal factors: functioning is more directly targeted and may improve rapidly, while QoL may require longer to emerge as functional changes accumulate. Most studies' short follow-up periods (approximately 3 months) may be insufficient to capture delayed QoL benefits. Third, adults with ADHD often hold negative self-schemas and maladaptive cognitions (Knouse et al., 2019; Philipsen et al., 2017), which may make them more critical when evaluating their overall life satisfaction, even when they experience objective functional improvements. Fourth, CBT skills are designed to target functioning domains (e.g., time management for improving academic or work performance). In contrast, QoL may be influenced by many factors beyond CBT's direct targets (e.g., financial situation, physical health), which are often impaired in adults with ADHD (French et al., 2024; Libutzki et al., 2024). Fifth, the high rate of psychiatric comorbidity may explain differential effects. CBT targeting ADHD may improve functioning without fully addressing QoL when depression, anxiety, or other conditions also contribute to reduced life satisfaction. This interpretation is tentatively supported by our meta-regression finding that higher comorbidity rates predicted poorer QoL outcomes. Taken together, these modest QoL effects are consistent with broader transdiagnostic CBT research showing similar small-to-moderate improvements across multiple conditions (Fordham et al., 2021), although the extent to which ADHD-specific versus transdiagnostic factors account for these patterns remains unclear.

SCDs provided case-level observations consistent with these findings. They also suggest that sustained improvements may depend on continued skill application (van Emmerik-van Oortmerssen et al., 2015), and that reduced engagement was associated with partial relapse (Puente & Mitchell, 2016). Although not directly tested, this pattern aligns with the within-subject findings and is consistent with the possibility that CBT may produce skill-dependent gains rather than stable trait changes, which may have implications for long-term skills maintenance, as supported by recent literature (López-Pinar et al., 2023). Detailed case-level data also suggested specific improvement mechanisms, such as enhanced organization and time management, which may be associated with better academic performance and social engagement (Eddy et al., 2015; López-Pinar & Vicente-Gispert, 2023).

4.2. CBT effect on specific functioning domains

Across specific domains, CBT showed statistically significant effects on all outcomes except for between-group academic functioning at follow-up. Among them, the most favorable quality of evidence was for social functioning outcomes. Although different instruments were used, studies generally assessed similar aspects of social functioning including interpersonal relationships, social activities/leisure participation, social support, and sexual life/intimacy. CBT produced small-to-moderate effects for social functioning with some evidence of maintenance at follow-up for both controlled and uncontrolled outcomes. These results are consistent with previous meta-analyses showing that CBT combined with medication may improve social functioning compared to pharmacotherapy alone (Li & Zhang, 2024), with the novel finding that these gains appear to be sustained at follow-up. The apparent stability of effects at follow-up could be particularly noteworthy given that social impairments are among the most distressing consequences of adult ADHD (Holst & Thorell, 2020), though this interpretation remains speculative. Our findings are consistent with other meta-analyses suggesting that CBT may improve social functioning in clinical populations (Hong et al., 2025), including other neurodevelopmental disorders (Yu

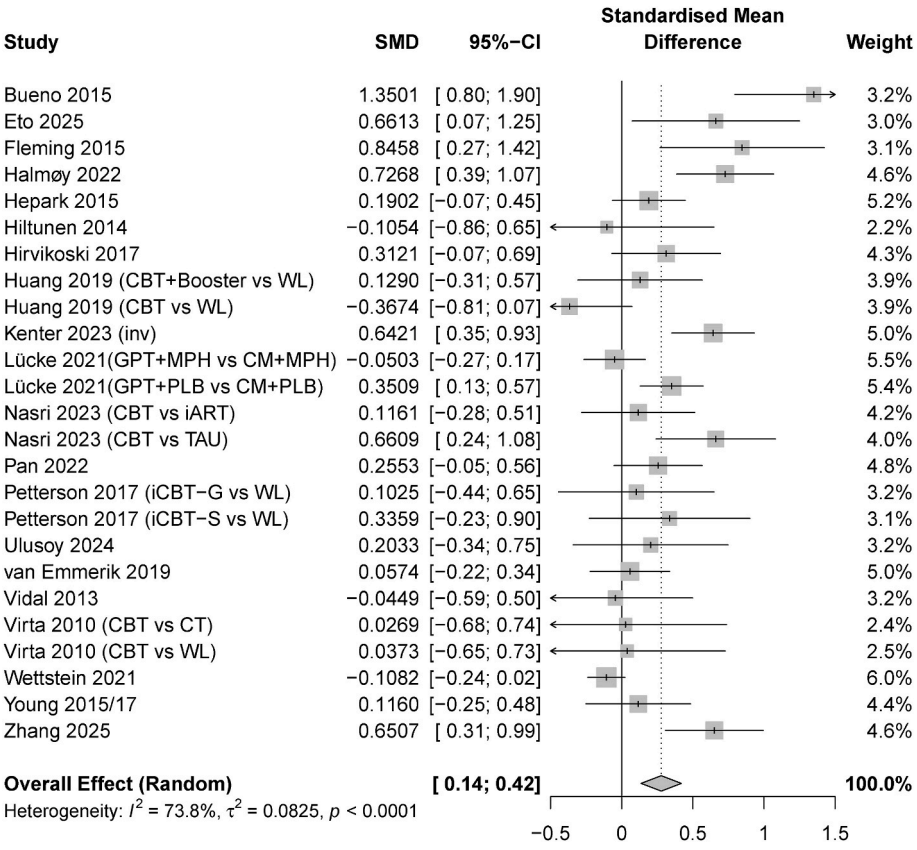


Fig. 3. Forest plot for between-group QoL at post-treatment.

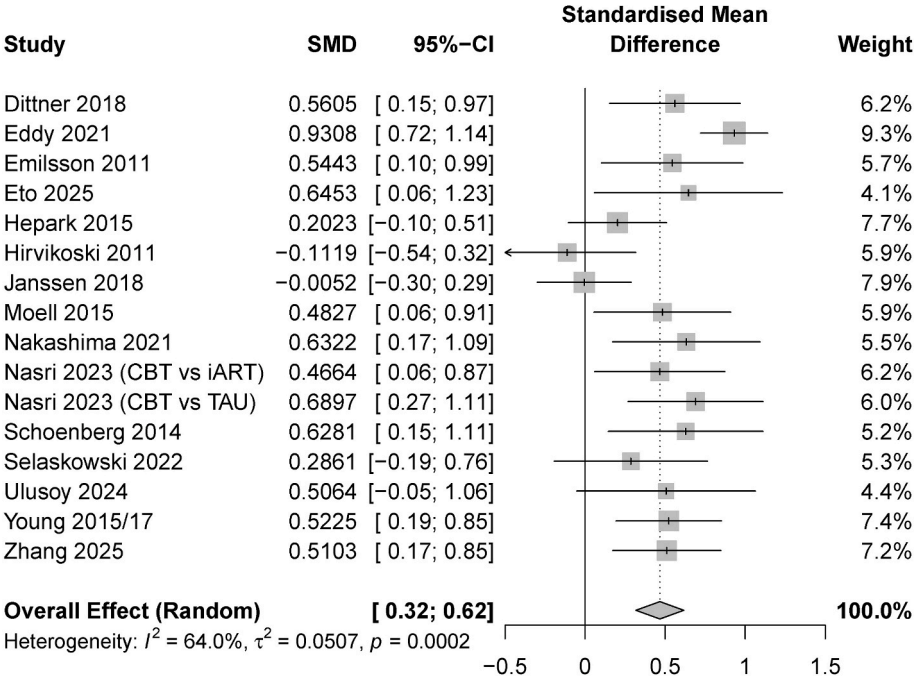


Fig. 4. Forest plot for within-subject QoL at post-treatment.

et al., 2023). Notably, CBT plus medication appeared superior to medication alone for improving social functioning in patients with depression (Zhang et al., 2016). Together, these findings are consistent with the possibility that CBT may facilitate the transfer of learned skills beyond the therapeutic setting. From a clinical perspective, this raises the possibility that CBT may address areas not fully targeted by pharmacotherapy, which primarily focuses on symptom control rather than teaching interpersonal skills or strategies for navigating social challenges.

The effects on occupational functioning were larger, with CBT

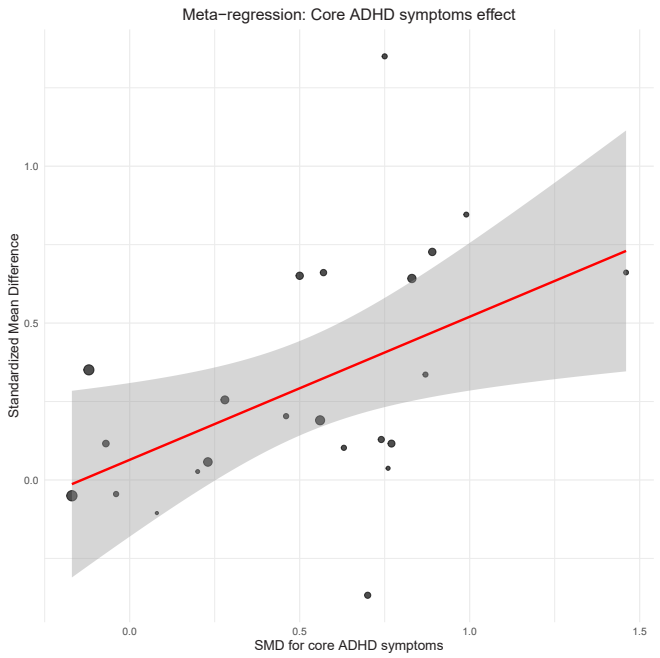


Fig. 5. Scatter plot for between-group QoL at post-treatment (left panel) and follow-up (right panel).

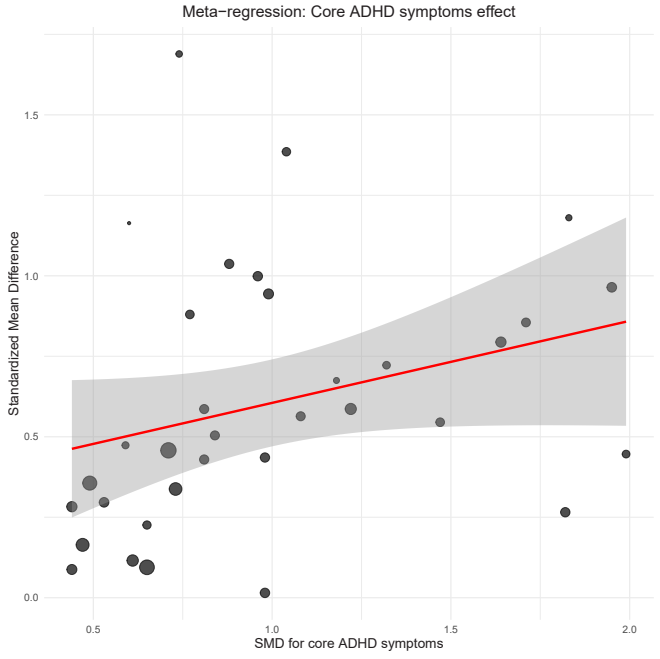


Fig. 6. Scatter plot for within-subject global functioning at post-treatment.

appearing to produce large improvements, with some evidence of persistence at follow-up. These estimates exceeded those reported in previous meta-analyses (Lauder et al., 2024). This may be due to prior reviews aggregating occupational outcomes with broader measures such as QoL and global functioning, which may dilute the domain-specific effect. However, the quality of the evidence for this outcome was moderate at post-treatment but decreased to low or very low at follow-up, primarily because of small sample sizes. Therefore, while these improvements appear promising, they remain preliminary and require confirmation in larger samples. These findings may have clinical relevance, given that employment-related impairment is often a primary concern for adults with ADHD seeking treatment (Kosheleff et al., 2023). Although these interpretations were not directly tested, the effects on work functioning could be consistent with CBT's emphasis on skills directly applicable to employment contexts, such as time management, planning, and breaking down complex tasks—skills that might theoretically address the executive function deficits that may impair workplace performance in adults with ADHD (Oscarsson et al., 2022). Furthermore, CBT's emphasis on environmental self-management strategies (e.g., workspace organization, minimizing distractions, implementing visual cues) may facilitate generalization and maintenance of learned skills to occupational settings, which may support the application of learned skills in occupational settings, and could be related to the observed effects at follow-up.

Finally, in contrast to social and occupational functioning, CBT showed more limited effects on academic functioning. Between-group effects disappeared at follow-up, within-subject effects were small, and the quality of evidence was low or very low. This was unexpected, since previous studies such as that by Eddy et al. (2021) found that study skills were significantly improved after CBT. The authors proposed that translating improved study skills into better grades may require more time. This more limited effect on academic functioning could also be due to the small sample sizes in the analyses and may be explained by the measurement approach: many included studies assessed academic functioning using GPA, which may not fully capture functioning in this domain. Previous studies also found no significant change in GPA after CBT interventions, despite improvements in other important outcomes (LaCount et al., 2015; Van der Oord et al., 2020). Furthermore, academic improvement may require multimodal interventions that extend beyond CBT's scope, including academic coaching, specialized tutoring, or assistive technologies. These domain-specific patterns may suggest that CBT effects on functioning vary depending on the extent to which intervention components target the behaviors required in each life context.

4.3. *Moderators of treatment effect*

Before examining specific moderators, it is important to note that subgroup analyses found no significant differences between studies using active versus inactive control groups. This is consistent with, though does not provide direct evidence of the involvement of specific therapeutic mechanisms. These findings align with meta-analytic evidence showing that CBT effects in other populations may not be fully explained by nonspecific factors alone (Cuijpers, Reijnders, & Huibers, 2019). The moderator patterns described below further inform understanding of which treatment characteristics may optimize these processes.

Regarding treatment characteristics, individual CBT showed significantly larger effects on global functioning than group interventions, consistent with previous ADHD reviews (López-Pinar et al., 2018), but contrasting with other populations (Cuijpers, Noma, et al., 2019). This could be because individual therapy may be better suited for ADHD's heterogeneous presentation (Nigg et al., 2020) through personalized problem-solving support for follow-through difficulties and by adjusting treatment pacing to each client's learning needs. In addition, remote CBT produced larger QoL effects, which is in line with preliminary evidence for online CBT in improving functional outcomes (Shou et al., 2022). This may reflect factors such as increased convenience and accessibility, as well as the possibility that delivering intervention within clients' everyday environments may facilitate skill implementation. However, functioning outcomes may ultimately require consistent skill application regardless of delivery format. Regarding the type of intervention, second-wave CBT outperformed mindfulness for functioning although the reasons for this difference were not directly examined. One speculative interpretation is that second-wave CBT may more explicitly target executive functioning skills (Ramsay, 2010), a primary concern for adults with ADHD, whereas mindfulness primarily addresses emotion

regulation and self-regulation. Moreover, more sessions predicted larger functional effects, suggesting a possible dose-response relationship allowing skill consolidation. This has been also previously described for CBT interventions in broader populations (Olatunji et al., 2014). Other intervention characteristics showed no significant moderating effects, including delivery format (therapist-led vs. self-delivered), type of concurrent medication, or the proportion of participants taking medication. Previous meta-analyses suggest that adding CBT to pharmacotherapy may produce additional small-to-moderate improvements in core ADHD symptoms in adults at post-treatment, though evidence for functional impairment and QoL remains limited and less consistent across age group (Li & Zhang, 2024). Our findings add that CBT may be effective regardless of participants' medication status. In addition, these null findings may suggest that CBT's effects on functioning and QoL do not substantially differ across the implementation contexts examined, although this interpretation should be considered cautiously.

Regarding participant characteristics, a higher proportion of male participants predicted larger functioning effects, possibly because women with ADHD have higher internalizing comorbidity rates (Siddiqui et al., 2024). A similar pattern has been observed for the effects of CBT on emotional comorbidities in adults with ADHD (López-Pinar et al., 2020), consistent with the possibility that higher comorbidity may interfere with the implementation of compensatory strategies and objective functional improvements. Conversely, higher comorbidity predicted larger QoL effects, possibly reflecting a floor effect where more impaired participants experienced greater subjective wellbeing improvements despite limited objective functional gains. Older age predicted larger effects on academic functioning, possibly reflecting greater maturity in implementing learned strategies. Finally, the effect was similar for participants with and without a formal ADHD diagnosis. This raises the possibility that CBT-based interventions targeting functional outcomes may also benefit individuals with subclinical ADHD presentations, a population frequently seen in clinical settings, though this interpretation warrants replication. However, given that these moderator analyses represent preliminary exploratory findings based on secondary variables, further research is needed to clarify these effects and their underlying mechanisms.

4.4. Meta-regressions examining relationships with ADHD symptom change

ADHD symptom improvement showed a stronger and more consistent association with QoL outcomes than with functioning outcomes, with significant relationships across most QoL measures and explaining considerable variance (22% to 68%). This relationship has also been observed for emotional comorbidities (López-Pinar et al., 2020). In contrast, functioning outcomes showed weaker relationships with symptom change, suggesting these improvements may occur more independently of symptom reduction. As previously discussed, this discrepancy may reflect the different nature of these measures, with QoL being more subjective and closely tied to symptom perception, while functioning may more directly reflect objective behavioral changes. Although these interpretations were not directly tested and remain speculative, the stronger association between symptom reduction and QoL could reflect shared influences such as enhanced self-efficacy and perceived control—key predictors of QoL (Bandura, 1997)—while functional improvements may rely more directly on implementing learned strategies, regardless of symptom perception. If replicated, this dissociation may suggest that symptom change and functional improvement may reflect partially independent psychological change processes.

4.5. Limitations

Several limitations should be considered when interpreting these findings. First, risk of bias was identified across study designs, although sensitivity analyses indicated that findings were robust when excluding high-risk studies, with effects remaining significant for all primary and most secondary outcomes. Second, publication bias was detected for several outcomes, suggesting potential overestimation of effects. Third, small sample sizes limited statistical power for specific functional domains at follow-up and constrained moderator analyses. Moreover, follow-up periods were typically short (3-6 months), restricting conclusions about lasting effects. Fourth, geographic overrepresentation from Western countries, particularly Sweden and the United States, limits generalizability to other cultural contexts where ADHD presentation, stigma, and treatment preferences may differ. Fifth, outcomes primarily relied on self-report measures, which, combined with the impossibility of blinding participants in psychotherapy trials, may inflate treatment effects through expectancy biases. Sixth, high heterogeneity across several outcomes indicates substantial unexplained variance, potentially reflecting unmeasured differences in outcome measures, treatment fidelity, or participant characteristics. In addition, the heterogeneity observed in forest plots, including near-zero and negative study-level effects, further cautions against overinterpreting pooled estimates as indicative of a consistent benefit across all settings and populations. Finally, the inclusion of single-arm trials in within-subject analyses limits causal inferences for these estimates; however, our primary conclusions are based on between-group comparisons.

4.6. Clinical implications

Despite these limitations, these findings may have potential implications for process-based models of behavioral change in adult ADHD, while offering tentative considerations for clinical intervention strategies. First, the results suggest that CBT may be associated with improvements in functional impairment (often a primary concern among treatment-seeking adults with ADHD) with some evidence of maintenance over time, although this should be interpreted cautiously given the heterogeneity and limited follow-up data. These findings are broadly consistent with the inclusion of CBT in treatment guidelines, even for patients already receiving medication, as CBT may address functional difficulties not fully resolved by medication alone. Second, in terms of intervention development, exploratory analyses suggest that individual formats and a higher number of sessions (approximately 12) may be associated with stronger functional outcomes compared to brief or group-based approaches, providing preliminary guidance for program design. Third, regarding service delivery, the observed effectiveness of CBT across different delivery formats (face-to-face and remote) and medication statuses may support its flexible implementation in contexts involving diverse patient needs and system-level constraints. Finally, future applied work should prioritize developing and testing strategies to enhance academic functioning outcomes and maintain QoL improvements over longer periods, while expanding access to underrepresented populations, particularly non-Western communities.

5. Conclusions

This meta-analysis found moderate-to-large effects of CBT on global functioning in adults with ADHD supported by promising but heterogeneous evidence, with some evidence of maintenance at follow-up. The absence of significant differences between active and inactive control conditions is consistent with, though does not directly demonstrate, the operation of specific therapeutic mechanisms. Domain-specific analyses revealed the strongest albeit preliminary effects for occupational functioning, apparent improvements in social functioning, and more limited benefits for academic functioning. Effects on self-reported QoL were smaller and less sustained. By quantitatively synthesizing 70 studies with 5152 participants, this review considerably expands the evidence base and provides the first comprehensive examination of domain-specific functioning outcomes.

However, methodological concerns warrant caution. Risk of bias was

identified across study designs, and geographic overrepresentation limits generalizability beyond Western populations. Future research should employ longer follow-up periods to capture potential delayed QoL benefits, incorporate objective functioning measures alongside self-report, and include more diverse samples to establish cross-cultural validity. Despite these limitations, the current findings suggest that CBT may represent a promising intervention for improving functioning in adults with ADHD, with potential implications for clinical practice and future intervention development.

CRediT authorship contribution statement

Carlos López-Pinar: Writing – original draft, Visualization, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Benjamin Selaskowski:** Methodology, Investigation, Data curation. **Marcel Schulze:** Writing – review & editing, Methodology, Investigation, Data curation. **Martin Oscarsson:** Writing – review & editing, Investigation, Data curation. **Javier Fenollar-Cortés:** Writing – review & editing. **Margaret H. Sibley:** Writing – review & editing. **Alexandra Philipsen:** Writing – review & editing. **Tatja Hirvikoski:** Writing – review & editing.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the authors used Claude (Anthropic) in order to improve language and readability of the manuscript text and to review and optimize R code for statistical analyses. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.brat.2026.105026.

Data availability

The data is freely available in the Open Science Framework: https://doi.org/10.17605/OSF.IO/AVKWT.

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